

Ian Schillebeeckx

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EDUCATION

Washington University in St. Louis

PhD, Computer Science (Machine Learning) – Fastest graduation in department history

St. Louis, MO

2012 – 2016

Saint Louis University

Honors BS, Computer Science

Honors BS, Applied Mathematics

St. Louis, MO

2007 – 2011

EXPERIENCE

Director of Enterprise Data Science and AI

January 2025 – Present

CareDx, Inc. (Diagnostics, ~750 employees)

San Francisco, CA

- Built company's first centralized DS function from zero; hired, developed, and retained high-performing team; partnered daily with Finance, Commercial, Operations, and Executive teams; served as quantitative thought leader for CEO, CFO, and COO, presenting board-level decisions backed by rigorous modeling
- Owned revenue data layer: built forecasting system covering ~75% of revenue with custom time series optimization (ARIMA, Prophet, custom objectives) across 3/6/12/24-month horizons, reducing error from 6% to 0.9%; Databricks ETL pipelines for model training and inference; weekly C-suite reports, quarterly/annual CFO planning, Wall Street reporting
- Led unit economics and causal analysis: applied causal DAGs, causal forests, DiD, and propensity matching to quantify marginal impact of sales and billing processes on revenue; translated findings into data-driven business rules driving 10% revenue increase (~\$25M)
- Directed churn/retention model covering \$270M revenue: novel piecewise linear changepoint detection + dynamic programming approach for transaction-like time series; automated anomaly detection with CI/CD; built LLM-based next-best-action agent generating personalized engagement recommendations per territory; customer segmentation insights drove proactive retention across at-risk accounts
- Shipped 3 LLM applications in production: enterprise RAG app (\$1M investment) for self-service knowledge retrieval; ProductLLM scoring product profiles against market potential rubric; PaperLLM generating original research manuscripts
- Operate AI-first: built custom Codex tool stack (database access, SQL generation, schema introspection) for data science team, automating recurring analytical pipelines and agentic workflows; taught Codex adoption across broader 20-person Data and AI group (Statistics, Data Engineering, Bioinformatics, Data Science) through executive sponsorship and regular peer training; drove company-wide Copilot adoption to 92% (~750 employees), saving 8% of total hours annually; personally matched output of 5-member team using agents

Director of Data Science

January 2024 – January 2025

CareDx, Inc.

San Francisco, CA

- Led multidisciplinary data organization (14 people, \$6M budget) spanning Bioinformatics, Statistics, and Data Science; managed managers; built multi-quarter roadmap with clear line-of-sight to business outcomes; set standards for modeling rigor, data quality, pipeline reliability, and production-readiness
- Directed production ML models (LSTM, SVM, LASSO) end-to-end from development through deployment, monitoring, and governance in regulated environment
- Oversaw enterprise analytics portfolio of ~20 Tableau dashboards serving Sales, Finance, Operations, and Clinical; set metric standards and data governance across the organization

Head of Product and Data Science

December 2017 – January 2024

Cofactor Genomics (Healthcare / Diagnostics Startup)

San Francisco, CA

- Owned end-to-end delivery of 2 production ML models: feature engineering, model optimization via bootstrap cross-validation, scalable cloud deployment with automated regression testing, CI/CD, and model monitoring
- Architected data platform from scratch: model repository (SageMaker), versioned training data (100s GBs), cloud pipelines (AWS Batch, +200% throughput, -25% cost), CI/CD (GitHub Actions), reconciliation pipelines ensuring data quality and accuracy at scale
- Designed and executed randomized controlled experimentation program across 12+ sites (~500 subjects): prespecified hypotheses, statistical power analysis, A/B testing methodology and cohort analysis informing pricing, positioning, and strategic decisions
- Led multidisciplinary team of 10 in high-growth B2B startup; defined multi-quarter roadmaps and OKRs; partnered with Engineering, Product, and Commercial on pricing, packaging, and market strategy

- First author on Nature ML paper (2nd most downloaded in 2022); authored 3 AI patents (2 granted)

CEO & Founder

May 2017 – August 2020

Where4Care (B2C Technology Startup)

St. Louis, MO

- Built self-serve product from zero: designed matching algorithm, full tech stack (Python/Django), cloud infrastructure; validated through 200-user study; hands-on end-to-end from product definition through production

SELECTED PUBLICATIONS & PATENTS

3 Artificial Intelligence Patents (2 granted) – Methods for prediction, classification, and data processing using ML/AI
ML-based Prediction Model – *Nature Scientific Reports, 2022* (2nd most downloaded paper that year)

12 Peer-Reviewed Publications – Applied ML, deep learning, experimentation (*CVPR, ECCV, Nature, ICCP, 3DV*)

TECHNICAL SKILLS

Revenue & Unit Economics: Revenue Forecasting (ARIMA, Prophet, Custom), Cost/Billing Data Models, Churn/Retention, Upsell/Expansion, LTV, Pricing & Packaging Optimization, Propensity Models, Reconciliation Pipelines, Exec Dashboards

Growth & Experimentation: PLG Data Science, Activation/Conversion Analytics, Funnel Analysis, A/B Testing, Causal Inference (DAGs, Causal Forests, DiD, Propensity Matching, Synthetic Controls), Incrementality Analysis

AI & Agentic Workflows: LLMs, RAG (Embeddings, Vector Search), Agentic Systems (Custom Tool Building, Codex), AI-Assisted Analytical Pipelines, Semantic Layer, AI-Native Team Operations

Languages & Platform: Python, SQL, R, AWS (SageMaker, Batch), Databricks, ETL/Data Pipelines, Tableau, Streamlit, CI/CD (GitHub Actions), Docker, Git

Leadership: Building & Running High-Performing DS Teams (up to 14 reports, \$6M budget), Managing Managers, Multi-Quarter Roadmap Execution, Executive Communication (C-Suite, Board), Cross-Functional Partnership (Finance, RevOps, Product, Engineering, Marketing)